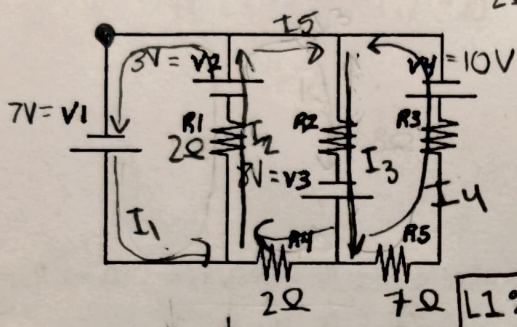


100% Excellent!

Name: _____

These questions assess your understanding of the material from Chapter(s) 20, 21 of OpenStax College Physics. Please write your name on the sheets of paper that you use. Carefully read each question. Credit is only awarded if you answer correctly and clearly show all major steps necessary to find your answer; credit is not awarded for missing work, unclear work, or the use of memorized equations absent from the equation sheet. Half credit may be awarded for insufficient work that suggests some degree of understanding. Half credit is awarded for correct numerical answers with incorrect or missing units. Correct numerical answers do not guarantee credit.

1. Solve for the power dissipated in resistor R_1 given $V_1 = 7\text{ V}$, $V_2 = 3\text{ V}$, $V_3 = 8\text{ V}$, $V_4 = 10\text{ V}$, $R_1 = 2\ \Omega$, $R_2 = 1\ \Omega$, $R_3 = 3\ \Omega$, $R_4 = 2\ \Omega$, and $R_5 = 7\ \Omega$.



$$= 2\ \Omega \quad p = \frac{V_2}{R}$$

$$\text{ANSWER: } 50\text{ W}$$

$$\Delta V = 0$$

$$V_1 - I_2 R_1 + V_2 - I_3 R_2 + V_3 - I_5 R_4 - I_4 R_5 - I_4 R_3 - V_4 = 0$$

$$7 - 2I_2 + 3 - I_3 + 8 - 2I_5 - 7I_4 - 3I_4 - 10 = 0$$

$$8 = 2I_2 + I_3 + 2I_5 + 10I_4$$

$$\text{L1: } 7 - 2I_2 + 3 = 0$$

$$10 = 2I_2$$

$$I_2 = 5$$

left-most loop

$$R_1 \text{ 's power dissipated}$$

$$P_1 = (I_2)^2 (R_1)$$

$$P_1 = (5)^2 (2) = 50\text{ W}$$

$$\text{L2: } V_2 - I_2 R_1 - I_3 R_2 + V_3 = 0$$

$$3 - 2I_2 - I_3 + 8 = 0$$

$$2I_2 - I_3 + 11$$

middle loop

$$I_2 = I_1 + I_5$$

$$I_3 = I_4 + I_5$$

2. A car headlight filament is made of tungsten ($\rho = 5.6 \times 10^{-8}\ \Omega \cdot \text{m}$) and has a resistance of $0.54\ \Omega$. If the filament is a cylinder 6.9-cm long, what is its diameter?

$$R = 0.54\ \Omega$$

$$L = 0.069\text{ m}$$

$$A = \frac{\rho L}{R}$$

$$A = \frac{(5.6 \times 10^{-8})(0.069)}{0.54} = 7.155 \times 10^{-9}\text{ m}^2 = \pi r^2$$

$$r = 4.7725 \times 10^{-5}\text{ m}$$

$$d = 9.545 \times 10^{-5}\text{ m}$$

$$\text{ANSWER: } 9.55 \times 10^{-5}\text{ m}$$

3. Suppose the voltage supplied to a circuit is 7.3 V. There are three resistors wired in series: $R_1 = 3.9\ \Omega$, $R_2 = 5.0\ \Omega$, and $R_3 = 1.0\ \Omega$. Calculate the power output of the battery.

$$V = 7.3\text{ V}$$

$$R_s = 3.9 + 5 + 1 = 9.9\ \Omega$$

$$P = \frac{(V)^2}{R} = \frac{(7.3)^2}{9.9} = 5.383\text{ W}$$

$$\text{ANSWER: } 5.383\text{ W}$$

+3

4. An incandescent lightbulb has a 1.0-cm-long filament and operates at 105. V to deliver 50. W. Calculate the diameter of the bulb's filament, assuming it to be a cylindrical tungsten wire, given that the resistivity of tungsten is $5.6 \times 10^{-8} \Omega \cdot \text{m}$.

$$L = 1 \text{ cm} = 0.01 \text{ m}$$

$$V = 105 \text{ V}$$

$$P = 50 \text{ W}$$

$$\rho = 5.6 \times 10^{-8} \Omega \cdot \text{m}$$

$$P = \frac{V^2}{R}$$

$$R = \frac{V^2}{P}$$

$$R = \frac{(105)^2}{50} = 220.5 \Omega$$

$$A = \frac{\rho L}{R}$$

$$A = \frac{(5.6 \times 10^{-8})(0.01)}{220.5 \Omega} = 2.54 \times 10^{-12} \text{ m}^2 = \pi r^2$$

$$r = 8.99 \times 10^{-7} \text{ m}$$

$$d = 1.798 \times 10^{-6} \text{ m}$$

$$\text{ANSWER: } 1.798 \times 10^{-6} \text{ m}$$

5. Suppose the voltage supplied to a circuit is 4.2 V. There are three resistors wired in series: $R_1 = 5.9 \Omega$, $R_2 = 2.9 \Omega$, and $R_3 = 7.3 \Omega$. Calculate the current in the circuit.

$$V = 4.2 \text{ V}$$

$$R_s = 5.9 + 2.9 + 7.3$$

$$R_s = 16.1 \Omega$$

$$V = IR$$

$$I = \frac{V}{R} = \frac{4.2 \text{ V}}{16.1 \Omega} = 0.261 \text{ A}$$

$$\text{ANSWER: } 0.261 \text{ A}$$

6. A 12.9-V car battery with internal resistance 0.012Ω supplies a starter motor drawing 190 A for 4.1 s. Calculate the electrical energy delivered to the starter during this time.

$$V = 12.9 \text{ V}$$

$$r = 0.012 \Omega$$

$$I = 190 \text{ A for } 4.1 \text{ s}$$

$$V = 12.9 - (190)(0.012)$$

$$V = 10.62 \text{ V}$$

$$P = IV = \frac{E}{t} = \frac{V^2}{R} = I^2 R$$

$$E = IVt$$

$$E = (190 \text{ A})(10.62 \text{ V})(4.1 \text{ s})$$

$$E = 8.27 \times 10^3 \text{ J}$$

$$8.27 \text{ E3}$$

$$\text{ANSWER: } 8.27 \times 10^3 \text{ J}$$

note that

$$P = \frac{E}{t} = I^2 R$$

can also be used (resistance of starter motor) (not given)

7. Calculate the power dissipated by a $2.2\text{-}\Omega$ car headlight that draws a current of 4.0 A .

ANSWER: 35.2 W

$$R = 2.2\text{ }\Omega$$

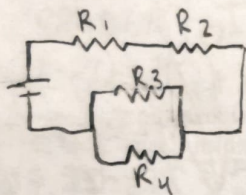
$$I = 4\text{ A}$$

$$P = I^2 \cdot R$$

$$P = (4)^2 \cdot 2.2 = 35.2\text{ W}$$

8. A circuit has two branches connected in parallel: branch A consists of resistors $R_1 = 6.5\text{ }\Omega$ and $R_2 = 3.9\text{ }\Omega$ in series; branch B consists of resistors $R_3 = 1.3\text{ }\Omega$ and $R_4 = 3.7\text{ }\Omega$ in parallel. Draw this circuit and find the total equivalent resistance. Note: you must draw the circuit correctly to receive credit.

ANSWER: $0.88\text{ }\Omega$



$$R_{12} = 6.5 + 3.9 = 10.4\text{ }\Omega$$

$$R_{34} = \left(\frac{1}{1.3} + \frac{1}{3.7} \right)^{-1} = 0.962\text{ }\Omega$$

$$R_{TOT} = \left(\frac{1}{10.4} + \frac{1}{0.962} \right)^{-1} = 0.88\text{ }\Omega$$

9. What current flows through a high-voltage transmission line to transmit $1.6 \times 10^3\text{ MW}$ of power at 540 kV ?

ANSWER: $2.96 \times 10^3\text{ A}$

$$P = 1.6 \times 10^3\text{ MW} = 1.6 \times 10^9\text{ W}$$

$$V = 540\text{ kV} = 540 \times 10^3\text{ V}$$

$$P = IV$$

$$I = \frac{P}{V} = \frac{(1.6\text{ E}9\text{ W})}{(540\text{ E}3\text{ V})} = 2.96 \times 10^3\text{ A}$$

MKHD
6321

+3

10. What is the resistance of an automobile headlight which carries a current of 1.4 A when 5.6 V is applied to it?

ANSWER: 4 Ω

$$I = 1.4 \text{ A}$$

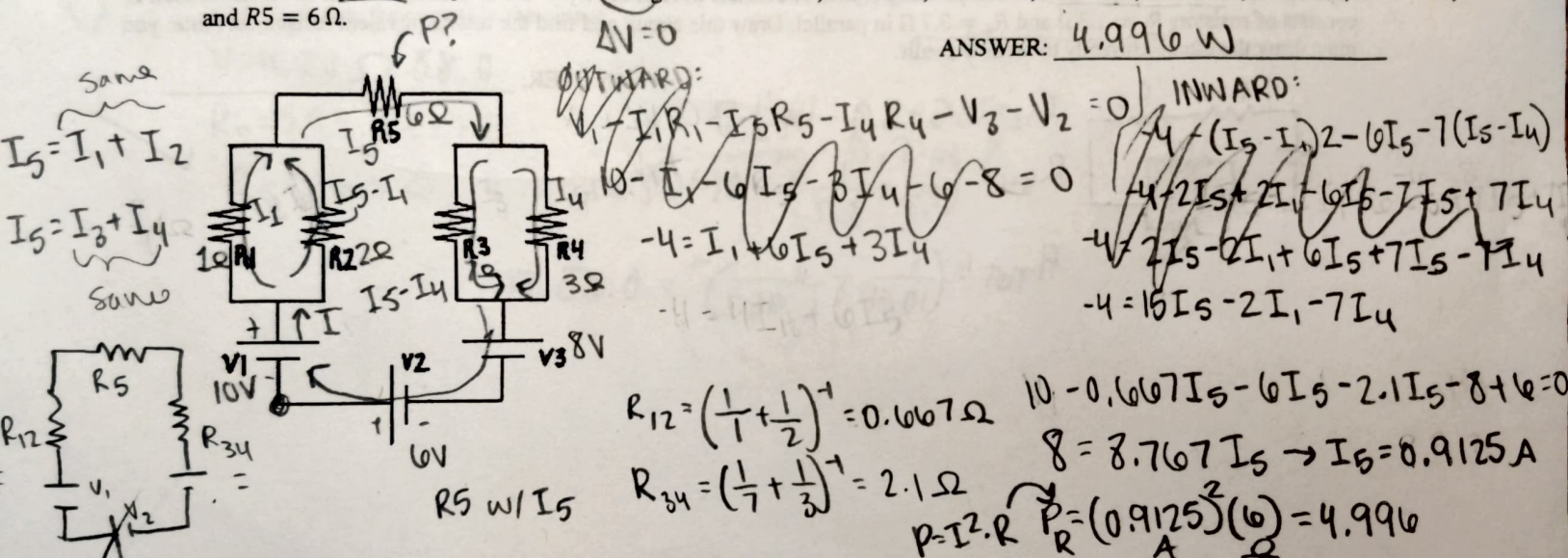
$$V = 5.6 \text{ V}$$

$$V = IR$$

$$R = \frac{V}{I} = \frac{5.6 \text{ V}}{1.4 \text{ A}} = 4 \Omega$$

11. Solve for the power dissipated in resistor R_5 given $V_1 = 10 \text{ V}$, $V_2 = 6 \text{ V}$, $V_3 = 8 \text{ V}$, $R_1 = 1 \Omega$, $R_2 = 2 \Omega$, $R_3 = 7 \Omega$, $R_4 = 3 \Omega$, and $R_5 = 6 \Omega$.

ANSWER: 4.996 W



12. A 11.9-V car battery supplies a starter motor drawing 620 A for 1.2 s. Calculate the total charge moved by the battery during this time.

ANSWER: 744 C

$$V = 11.9 \text{ V} \quad \text{Find: } \Delta Q$$

$$I = 620 \text{ A}$$

$$t = 1.2 \text{ s}$$

$$I = \frac{\Delta Q}{\Delta t}$$

$$\Delta Q = \Delta t \cdot I$$

$$= 1.2 \cdot 620$$

$$= 744 \text{ C}$$

+3

PHY202 Exam 3 on chapter(s): 20, 21

1. $P = 50.0 \text{ W} \therefore$ Kirchhoff's rules and $P = I^2 R$
(Chapter 21, Section (3) Kirchhoff's Rules - tkd_circ2)
2. $D = 9.55 \times 10^{-5} \text{ m} \therefore R = \frac{\rho L}{A}$
(Chapter 20, Section (3) Resistance and Resistivity - cp1e_ch20_ex5)
3. $P = 5.38 \text{ W} \therefore R_s = R_1 + R_2 + R_3, V = IR, P = IV$
(Chapter 21, Section (1) Resistors in Series and Parallel - cp1e_ch21_ex2)
4. $d = 1.80 \times 10^{-6} \text{ m} \therefore P = \frac{V^2}{R}, R = \frac{\rho L}{A}$ and $A = \pi r^2$
(Chapter 20, Section (4) Electric Power and Energy - tkd2)
5. $I = 0.261 \text{ A} \therefore R_s = R_1 + R_2 + R_3, V = IR$
(Chapter 21, Section (1) Resistors in Series and Parallel - cp1e_ch21_ex1)
6. $E = 8.27 \times 10^3 \text{ J} \therefore P = IV = \frac{E}{t}, V = \text{emf} - Ir$
(Chapter 21, Section (2) Electromotive Force: Terminal Voltage - cp1e_ch21_ex4)
7. $P = 35.2 \text{ W} \therefore P = I^2 R$
(Chapter 20, Section (4) Electric Power and Energy - cp1e_ch20_ex7)
8. $R_{eq} = 0.881 \Omega \therefore R_s = R_1 + R_2 + R_3, \frac{1}{R_p} = \frac{1}{R_1} + \dots + \frac{1}{R_n}$
(Chapter 21, Section (1) Resistors in Series and Parallel - cp1e_ch21_ex3)
9. $I_{RMS} = 2.96 \times 10^3 \text{ A} \therefore P_{ave} = I_{RMS} V_{RMS}$
(Chapter 20, Section (5) Alternating Current v. Direct Current - cp1e_ch20_ex10)
10. $R = 4.00 \Omega \therefore V = IR$
(Chapter 20, Section (2) Ohm's Law - cp1e_ch20_ex4)
11. $P = 5.00 \text{ W} \therefore$ Kirchhoff's rules and $P = I^2 R$
(Chapter 21, Section (3) Kirchhoff's Rules - tkd_circ1)
12. $Q = 744. \text{ C} \therefore I = \frac{\Delta Q}{\Delta t}$
(Chapter 20, Section (1) Current - tkd0)